What the additional outcome identifies

Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). November 7, 2026 · Port Coquitlam, British Columbia, Canada.

> The orthogonal outcome identifies purity along the submitted interpolation path. Removing that restriction gives an exact interval. Even the additional first-qubit Z reading from a product input leaves the whole interval attainable at the values predicted along the path.

The normalized Choi ket $|M\rangle$ is $(M\otimes I)|\Phi_4\rangle$ for the four-dimensional input and output. Use $A=Z\otimes I$, $B=X\otimes I$ and the orthonormal kets $p=|I\rangle$, $u=|iA\rangle$, $v=|iB\rangle$. Fix the input probabilities $s=131/320$, $h=0.99$ and $q=81711338569043/224995905216000$. Put $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]=0.115482549888\ldots$, $d=s+k$, $r=1-d=0.475142450112\ldots$, and $w=\sqrt{s}\,p-\sqrt{k}\,u$. The fixed binary detector projects onto $e=\sqrt{h}\,p+\sqrt{(1-h)}u$. The second outcome is $F=|v\rangle\langle v|$.

The minimum-purity endpoint has weight d on the unitary Choi ket $w/\sqrt{d}$, and weight $r/14$ on each of the 14 remaining orthogonal Pauli kets. The pure endpoint $w+\sqrt{r}\,v$ corresponds to a unitary by anticommutation of A and B. Both channels and every convex mixture are mixed-unitary, trace-preserving and unital.

Quantity for the two named endpoints	Recomputed value
Choi half-trace distance	0.766538917119
Optimal one-copy equal-prior success	0.883269458559
Three-outcome total variation	0.441203703675
Product-input Z-positive probabilities	0.728490028508 and 0.524857549888

In the basis $w/\sqrt{d},v$, the Choi difference has diagonals $0,13r/14$ and off-diagonal $\sqrt{(rd)}$. The other 13 spectator eigenvalues are $-r/14$, and there is one zero direction. The resulting trace norm gives the quoted Helstrom success for these two Choi states. Optimization over all channel inputs is a separate discrimination problem.

The sharp interval after observing F

The premises here are the particular complete lower-endpoint phase curve, the known match population s, and the fully specified detector E above. The ordinary rate alone would supply less information. Write f=Tr(Fρ). The full curve fixes the compression of ρ onto span{p,u} to |w⟩⟨w|. Indeed its mean gives sh+(1−h)⟨u|ρ|u⟩, and the two quadratures fix the complex off-diagonal entry. Positivity eliminates the direction in this span orthogonal to w. The complementary subspace has population r, so 0≤f≤r.

> For each feasible f, the exact interval of normalized Choi purities is [J(f),1], with J(f)=d²+f²+(r−f)²/13. Every value occurs even within mixed-unitary channels.

Let D be the projection onto the 13-dimensional space perpendicular to p,u,v. In the orthogonal decomposition along w, v, and this remaining space, discarding off-diagonal blocks lowers the squared Hilbert–Schmidt norm. The last block has trace r−f, giving Tr(ρ²)≥J(f) by Cauchy–Schwarz. The state ρ(f)=|w⟩⟨w|+fF+(r−f)D/13 attains it. Its spectral terms are normalized unitary Choi kets with nonnegative weights, using the 13 Pauli operators other than I,A,B.

With L=d²+r²/14=0.291601186810187…, the lower endpoint is L+(14/13)(f−r/14)². In contrast, the submitted path gives L+(1−L)[(14f/r−1)/13]², and that formula applies only for r/14≤f≤r on that path.

Extra probability f	Minimum possible purity	Maximum possible purity
0	0.292841628283	1
r/14 = 0.03393875…	0.291601186810	1
r/2 = 0.23757123…	0.336257079801	1
r = 0.47514245…	0.501235795573	1

The maximum and all intermediate values are attained by the following construction. It preserves not only the full phase curve and f but also the Z-positive rate d+6(r−f)/13 for the product-input test. The latter is exactly the value associated with f on the submitted interpolation path.

Why the product-input test can stay blind

The minimum-purity state on page 2 gives $p_+=d+6(r-f)/13$ for input $|00\rangle$ and first-qubit Z readout. Its w component preserves the first-qubit value and its v component flips it. Among the other 13 tensor Paulis, six preserve that value. The supplied interpolation has the same formula since both p_+ and f are affine in its mixing parameter and the formula holds at its two endpoints.

A pure channel with these same probabilities

Use first-qubit Pauli matrices X and Z. Let $U=(\sqrt{s}\,I-i\sqrt{k}\,Z+i\sqrt{f}\,X)/\sqrt{(d+f)}$. Anticommutation gives $U\dagger U=I$. Choose a Hermitian unitary N on the first qubit such that $|\langle 0|UN|0\rangle|^2=6/13$. Such an N exists: apply U† to $\sqrt{(6/13)}|0\rangle+\sqrt{(7/13)}|1\rangle$ and choose its overall phase so its first entry a is real. For resulting entries (a,b), take N with rows (a,conj(b)) and (b,−a).

> The four-dimensional operator $V=(U\otimes I)[\sqrt{(d+f)}I_4+i\sqrt{(r-f)}(N\otimes X)]$ is unitary because $N\otimes X$ is a Hermitian involution and $d+r=1$.

The part of $|V\rangle$ with second-qubit factor I is $w+\sqrt{f}\,v$. The rest has second factor X and is orthogonal to p,u,v. Its ordinary phase curve and f are therefore correct. Applied to $|00\rangle$, the two terms of V end in orthogonal second-qubit states. The Z-positive probability is $d+(r-f)|\langle 0|UN|0\rangle|^2$, the required value.

Write $\rho_1=|V\rangle\langle V|$. Since the two states agree on the w population d, the v population f, and the remaining total r−f, $\mathrm{Tr}[\rho(f)\rho_1]=J(f)$. Therefore the mixture $(1-t)\rho(f)+t\rho_1$ has purity $J(f)+(1-J(f))t^2$ for $0\le t\le 1$. Its outcome probabilities and both Choi marginals are fixed. This proves both attainability and completeness. It does not give the same interval for arbitrary independent values of f and p_+.

For exact s,h and phase curve, an interval of admissible f values yields a purity lower bound by minimizing J over that interval intersected with [0,r]. Real uncertainty in the other inputs also has to be propagated. Plugging a measured frequency into an exact equality would not do so.

A local check at the upper F rate

Consider the exact boundary f=r. There is no population outside span{η,v}, η=w/√d. Positivity forces ρ to have the two-by-two form with diagonal entries d,r and off-diagonal γ, $|\gamma| \le \sqrt{(dr)}$. This yields $\mathrm{Tr}(\rho^2)=d^2+r^2+2|\gamma|^2$. The Choi marginal $I_4/4$ forces γ to be real: the two operators giving η,v are W=(√s I−i√k A)/√d and iB; W†(iB) is a nonzero traceless anti-Hermitian operator. In the partial trace the cross term therefore vanishes exactly for real γ.

If the same channel transfers to the product preparation |00⟩, W and iB give the first-qubit states ξ=(√s−i√k)|0⟩/√d and ν=i|1⟩, respectively. Their second-qubit state is |0⟩ in both cases. The Z probability is d for every γ. A calibrated first-qubit measurement of $O=|\xi\rangle\langle\nu|+|\nu\rangle\langle\xi|$ instead has expectation 2γ. Hence

> At f=r, with the specified exact phase curve and justified channel transfer, $\mathrm{Tr}(\rho^2)=d^2+r^2+\langle O\rangle^2/2$. This is a single transverse spin component with known orientation.

This last identity uses exact support saturation. A finite estimate close to r requires bounds on the residual population and calibration errors before using a two-dimensional reduction. None of the F, Z or transverse outcomes here are measured hardware values.

A physical requirement on conditioning

For a nonzero trace-nonincreasing instrument branch B, define H=B†(I). The input success probability on state τ is Tr(Hτ). The input marginal of its normalized Choi state is $H^{T}/\mathrm{Tr}\,H$, so it equals $I_4/4$ precisely when H is proportional to I. Testing this property uses characterized preparations. If the history precedes the new input, common subsequent dynamics across preparations is a distinct transfer question. White et al. demonstrate experimental characterization of multitime processes with interventions.

Keith et al., Physical Review A 98, 042318 (2018), incomplete measurement and compatible states.

Watrous, The Theory of Quantum Information, chs. 3–4, state discrimination and mixed-unitary channels.

White et al., PRX Quantum 3, 020344 (2022), non-Markovian quantum process tomography.